which can be transformed into fissile ones, i.e. $^{232}$Th and $^{238}$U, which through neutron capture and β-decays form fissile $^{233}$U and $^{239}$Pu, respectively. The chemical reprocessing removes the fission products and actinides other than U and Pu. Some of the removed elements might be valuable enough to be isolated although this is seldom done. The mixed fission products and waste actinides are stored as *radioactive waste*. The recovered fissile materials may be re-fabricated (the U may require re-enrichment) into new elements for reuse. This "back-end" of the nuclear fuel cycle is discussed in Chapter 21.

## 19.2. ENERGY RELEASE IN FISSION

From Table 11.1 it is seen that in thermal neutron fission of $^{235}$U the fission fragments are released with a kinetic energy of ∼165 MeV (on the average), the 2.5 *prompt neutrons* have an average kinetic energy of ∼5 MeV together, and the *prompt γ-rays* have an average of 7 MeV. This *prompt energy release* of ∼177 MeV is absorbed in the surrounding material.

The fission products are radioactive and decay through emission of $\beta^-$, γ, and X-rays; their total amount of decay energy is ∼23 MeV. About 10 MeV (the value is uncertain) escapes the reactor as radiation, and ∼1 MeV of the decay energy remains as undecayed fission products in the spent fuel when unloaded from the reactor; thus ∼12 MeV βγ decay energy (divided about equally between $\bar{E}_\beta$ and $E_\gamma$) is absorbed in the reactor. The neutrons not consumed by fission are captured in the reactor material with release of binding energy; it is estimated that this amounts to about 10 MeV. Thus the total amount of energy expended per fission in a shielded controlled reactor is about $177 + 12 + 10 \simeq 199$ MeV. The total energy release varies only a few MeV between the different fissile nuclei. As a practical average value 200 MeV per fission can be used regardless of the fissioning nucleus:

$$E_f \approx 200 \text{ MeV per fission} = 3.20 \times 10^{-11} \text{ J per fission} \tag{19.2}$$

Thus $(3.20 \times 10^{-11})^{-1} \approx 3.1 \times 10^{10}$ fissions $s^{-1}$ correspond to the production of 1 W of reactor heat. The heat power of a reactor can be written

$$P = E_f \mathrm{d}N_f/\mathrm{d}t \tag{19.3}$$

where $\mathrm{d}N_f/\mathrm{d}t$ is the number of fissions per second. A nuclear power station producing 3 GW heat ($GW_{th}$; th for thermal) has an electric output of 1 $GW_e$ at a 33% *efficiency* in converting the thermal energy into electric; this corresponds to $8.1 \times 10^{24}$ fissions $d^{-1}$. Since the weight of a $^{235}$U atom is $M/\mathbf{N}_A = 3.90 \times 10^{-25}$ kg, this would correspond to the fission of $8.1 \times 10^{24} \times 3.9 \times 10^{-25} = 3.16$ kg $^{235}$U $d^{-1}$; the real consumption of $^{235}$U is slightly different and varies with time as discussed in §19.9.

## 19.3. FISSION PROBABILITY

When uranium or plutonium is irradiated by neutrons, in addition to neutron capture followed by fission (n,f ), several different processes occur: scattering, (n,γ), (n,2n) reactions, etc. All these reactions are of importance for the reactor designer as well as for the chemists who have to manufacture new fuel pellets and take care of the spent fuel elements. The probability for the various reactions depends on the neutron energy. As is seen in Figure 19.3, three regions are clearly distinguishable:

**(1)** For thermal neutrons with average kinetic energies ($\bar{E}_n$) $\leq$1 eV, fission of $^{235}$U and $^{239}$Pu dominates over neutron capture (i.e. $\sigma_f > \sigma_{n,\gamma}$). Though $\sigma_{n,\gamma}$ for $^{238}$U is small considerable capture occurs because of the large fraction of $^{238}$U normally present.

**(2)** For epithermal neutrons ($1 \leq \bar{E}_n \leq 10^5$ eV) large radiative capture and fission resonances occur. In this region heavier isotopes are formed by (n,γ) reactions: $^{235}$U(n,γ)$^{236}$U; $^{238}$U(n,γ)$^{239}$U; $^{239}$Pu(n,γ)$^{240}$Pu; $^{240}$Pu(n,γ)$^{241}$Pu; etc (see Fig. 19.5).

**(3)** For fast neutrons ($\bar{E}_n \geq 0.1$ MeV) the cross sections are relatively small, $\leq$1 b. Fission dominates over radiative capture. Of particular importance is that $^{238}$U becomes fissionable at a neutron energy of ~0.6 MeV; its fission cross section increases with neutron energy above the threshold to a constant value of ~0.5 b at $\geq$2 MeV.

It is obvious that the neutron energy spectrum of a reactor plays an essential role. Figure 19.4 shows the prompt (unmoderated) fission neutron spectrum with $\bar{E}_n$ ~2 MeV. In a nuclear explosive device almost all fission is caused by fast neutrons.

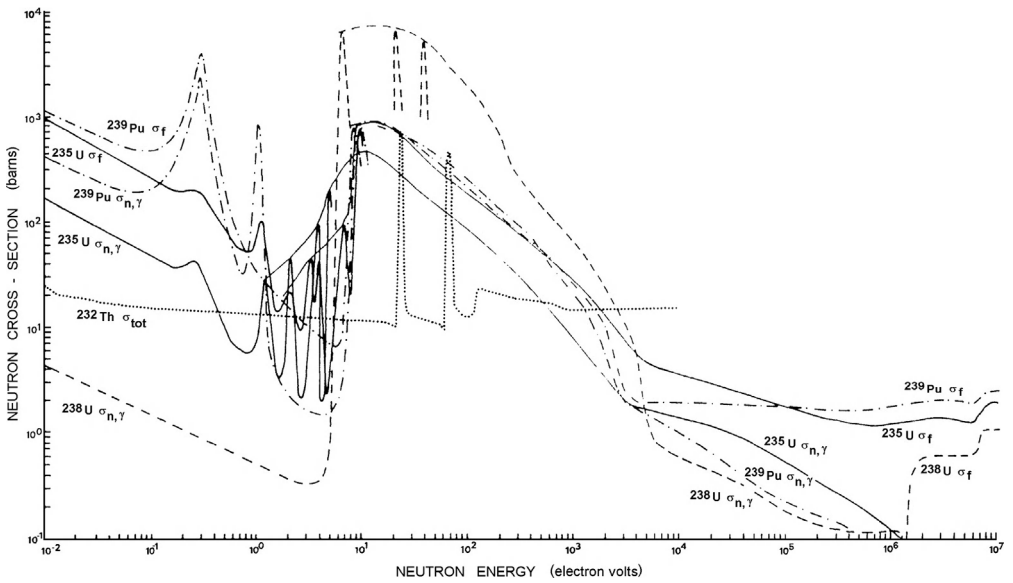



**Figure 19.3** Cross sections for n-capture ($\sigma_{n,\gamma}$), fission ($\sigma_f$), and total ($\sigma_{tot}$) as a function of neutron energy.